BIRZEIT UNIVERSITY

Flight Delay Prediction and Airport Punctuality Analysis Across Five European Countries

Phase 4 — Final Model and Discussion

Prepared By:

“Predictive Minds”

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Abstract

This final stage transforms the airport days dataset, collected in **phase** 1–3, into a working prediction model. we trained a logistic regression classifier to predict whether a major European airport will experience a delayed or on-time day, based on the airport's congestion level, type, season, day of the week, and year. The model achieves approximately 85% accuracy on unseen test data without any over-customization. It identifies on-time days almost perfectly and is fairly accurate regarding delays, but because delayed days represent a minority category, it still misses about 40%. The model's logic is fully explainable and consistent with the patterns found in our previous analysis: Lisbon and busy days lead to delays, while Dublin and Vienna are generally on-time, with a moderate summer peak.

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1. From analysis to prediction

Phases 1 through 3 produced detailed daily datasets for five major airports (Dublin, Copenhagen, Lisbon, Brussels, and Vienna) covering the period from 2023 to 2025. Phase 4 uses this data to answer a practical question: based on the information we have about a particular airport on a given day, can we predict whether it will experience delays that day? Because the course focused on binary classification, we framed the task as a binary classification problem. A day is classified as delayed (1) when the average air traffic management delay is at least one minute per arriving flight, and on time (0) otherwise. This one-minute minimum is the same threshold that separated the lowest on-time category in Phase 3, so the objective is consistent with the rest of the project.

Class balance

27% of airport-days are delayed and 73% are on-time. The classes are imbalanced, which is why we report precision, recall and F1 for the delayed class and do not rely on accuracy alone.

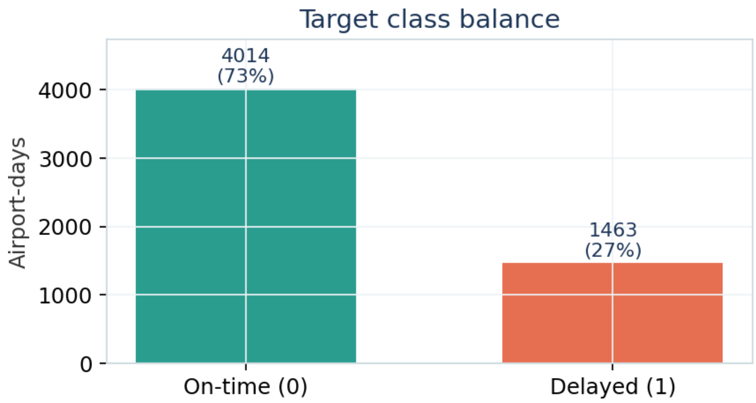


Figure 1. Distribution of the target across all 5,477 airport-days.

2. Features and the data-leakage problem

The most important decision in the model design was to avoid data leakage. Our goal is derived from delay columns, so re-entering any delay-derived column as a feature would allow the model to read its own answer, rendering the evaluation meaningless. Therefore, we excluded all delay-derived columns. We also removed columns that didn't indicate anything useful: airport type and scheduled service flag (consistent across all rows); latitude, longitude, and altitude (consistent within each airport, and thus redundant with its identity); the total flights column (representing the sum of departures and arrivals); and duplicate identity columns, retaining only one field for the country to represent the airport.

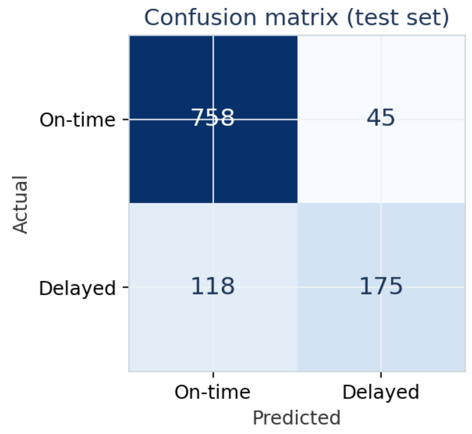
The final model uses only information that could be known before the day's delays: scheduled traffic volume (departures and arrivals), country/airport, season, day of the week, and year. We deliberately coded season as a category rather than a numerical month because logistic regression is linear, and a numerical month would incorrectly treat December and January (both winter) as separate.

3. Modelling method

We split the data into 80% for training and 20% for testing, using student ID (1222212) as a random seed to ensure reproducibility, and stratified the distribution based on the objective to maintain a 27% delay rate in both groups. Numerical values were compensated using the median and standardized, while categorical values were coded using single-hot encoding. Preprocessing and classifying were combined into a single path so that the transformers were applied only to the training data, preventing data leakage from the test set. The classifier is logistic regression, the binary model explained in the course: it passes a weighted sum of characteristics through a sigmoid function to calculate the probability of a delay of a given day.Results

Dataset	Accuracy	Precision	Recall	F1-score
Training	0.842	0.774	0.574	0.659
Test	0.851	0.795	0.597	0.682

Table 1. Overall performance. Training and test scores are almost identical, so the model is not over-fitting.



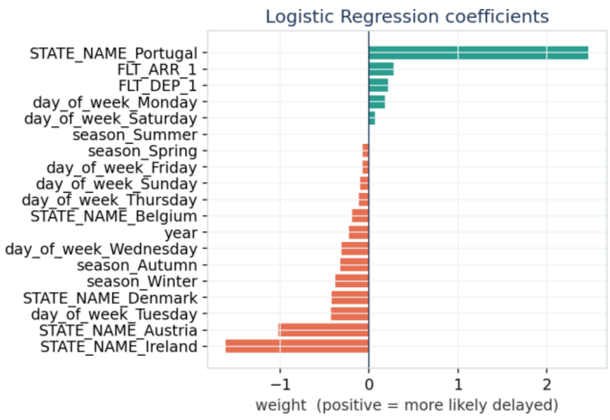


Figure 2 (left). Test-set confusion matrix. Figure 3 (right). Model coefficients — teal raises the chance of delay, orange lowers it.

On-time delivery days are identified with perfect accuracy (0.87 accuracy, 0.94 callback, F1 0.90 over 803 days). Delay days are more difficult to identify (0.80 accuracy, 0.60 callback, F1 0.68 over 293 days): when the model predicts a delay, it is usually correct, but it still classifies 118 out of 293 delayed days as on-time delivery days. These false negatives, reflected in the confusion matrix, are the model's main weakness.Discussion

The model is interpretable. Its coefficients (Figure 3) are consistent with the exploratory analysis from Phase III. Being in Portugal shows the highest positive weight, reflecting the high rate of delays in Lisbon, while Ireland and Austria show the highest negative weight, reflecting the punctuality of Dublin and Vienna. Traffic congestion increases the probability of a delay, which aligns with the intuition that traffic congestion causes delays, with summer being the most likely season. Therefore, the model is not a black box; it embodies logical and verifiable relationships.

The imbalance is significant. An accuracy of 85% seems good, but a model that simply guesses you will always arrive on time would achieve an accuracy of around 73%. The real telltale sign is the delay rating of 0.60: the model is cautious in identifying delays, so it misses some cases. For this reason, the methodology is not recommended to rely on the accuracy of unbalanced data.

This is a significant flaw. 85% accuracy seems good, but a model that simply assumes you will always arrive on time would achieve closer accuracy to 73%. The true indicator is a delay of 0.60: the model is careful in identifying delays, so it misses some. For this reason, the methodology advises against relying on the accuracy of this unbalanced data

4. Conclusion, limitations and future work

We developed an interpretable logistic regression model that accurately predicts flight times at airports throughout the day with a test accuracy of approximately 85% without any over-allocation. Even a simple model is useful, as it indicates airport, traffic, and seasonal combinations that are likely to cause delays—information that can support staffing and flight scheduling.

Limitations: The dataset covers only five major airports, so the model is highly dependent on airport-specific effects and may not be applicable to other airports. The target reflects arrival delays as determined by traffic management only. We were unable to use weather as a predictor because the weather columns in this dataset are weather-related delay minutes, which are derived from the delays themselves and could lead to data lea